

Hands-on Lab: Composing and Running Basic SQL Queries



IBM Developer
SKILLS NETWORK

Objectives

At the end of this lab you will be able to

- Compose and run basic queries

Exercise

So far in this module you learned about the five basic SQL statements to create tables, insert data, select results, update, and delete data. In this lab, you will practice composing and running these statements via hands-on experiences.

To write and execute the SQL statements in this lab, you will need to launch the SQL editor (Run SQL) in your Db2 database instance on IBM Cloud that you created in a previous lab:

- [Hands-on Lab : Sign up for IBM Cloud, Create Db2 service instance and Get started with the Db2 console.](#)

Now let's practice creating and running some SQL queries in the SQL editor.

Consider the following table called **INSTRUCTOR**:

ins_id	lastname	firstname	city	country
1	Ahuja	Rav	Toronto	CA
2	Chong	Raul	Toronto	CA
3	Vasudevan	Hima	Chicago	US

Task 0: Drop the table INSTRUCTOR from the database in case it already exists, so that we start from a clean state.

(Hint: Ignore the undefined error if this table does not already exist in your database)

Task 1: Create the INSTRUCTOR table as defined above. Have the ins_id be the primary key, and ensure the lastname and firstname are not null.

(Hint: ins_id is of type INTEGER, country of type CHAR(2), and rest of the fields VARCHAR)

Task 2A: Insert one row into the INSTRUCTOR table for the the instructor Rav Ahuja.

(Hint: values for the character fields require a singe quotation mark (‘) before and after each value)

Task 2B: Insert two rows at once in the INSTRUCTOR table for instructors Raul Chong and Hima Vasudevan.

(Hint: list the values for the second row after the first row)

Task 3: Select all rows from the INSTRUCTOR table.

Task 3B: Select the firstname, lastname and country where the city is Toronto

Task 4: Update the row for Rav Ahuja and change his city to Markham.

Task 5: Delete the row for Raul Chong from the table.

Task 5B: Retrieve all rows in the INSTRUCTOR table

Solution Script

Please try the above tasks on your own and only use the solutions if you need a hint or want to verify your queries.

If you would like to run all the solution queries of the SQL problems of this lab with a script, download the script below. Upload the script to the Db2 console and run. Follow [Hands-on Lab : Create tables using SQL scripts and Load data into tables](#) on how to upload a script to Db2 console and run it.

- [Solution-Script.sql](#)

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Changelog

Date	Version	Changed by	Change Description
2023-05-10	2.2	Eric Hao & Vladislav Boyko	Updated Page Frames
2021-03-10	2.1	Sandip Saha Joy	Created revised minor version
2020-08-31	2.0	Malika Singla	Markdown file creation

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